Paper Review • arXiv 2611.01560

Open Models Can Use Robots Well

uni-robot-agent: a unified, efficient, modular harness for robot-use agents

Democratizing Robot-Use Agents: A Unified Harness with Open Models

In one sentence: Without privileged information such as depth or task checkers, open Qwen3.8-Flash-Next with uni-robot-agent scores 31.39 on RoboDojo, beating the official GPT-6 Astra result (28.97); on 19 real-robot tasks it matches fine-tuned π0.5.

Tsinghua University • UC Berkeley | First submitted 2026-11-01

TL;DR

The Paper in Three Numbers

RoboDojo Score

31.39

vs 28.97

Open Qwen3.8-Flash-Next + our harness beats official GPT-6 Astra + RoboProbe; no depth, no task checker

Total score, 19 real tasks

49.50

vs 47.75

Zero-shot Qwen matches π0.5 fine-tuned on 10 h of demos per task; beats Dream-1 (35.50)

API cost per episode

≈19.5×

$21.29 vs $1.09

GPT-6 Astra scores higher (51.90) but costs ~19.5× more than Qwen

Core claim Well-structured information (camera calibration, robot description) plus reliable, feedback-rich tools suffices for open models to use robots effectively; this relies on combining visual understanding, spatial grounding, geometric coding and embodied reasoning, which future RUA training should improve jointly.

Motivation

Closed-model reliance limits RUA research to harness tuning, not training

1

No weights → no training

Research stays at harness design; cannot explore how further training improves RUA

2

API reliance → costly, slow

Large-scale interaction is costly; no model-level tricks like caching, yet inference speed is the main RUA bottleneck

3

Open-model patches → not transferable

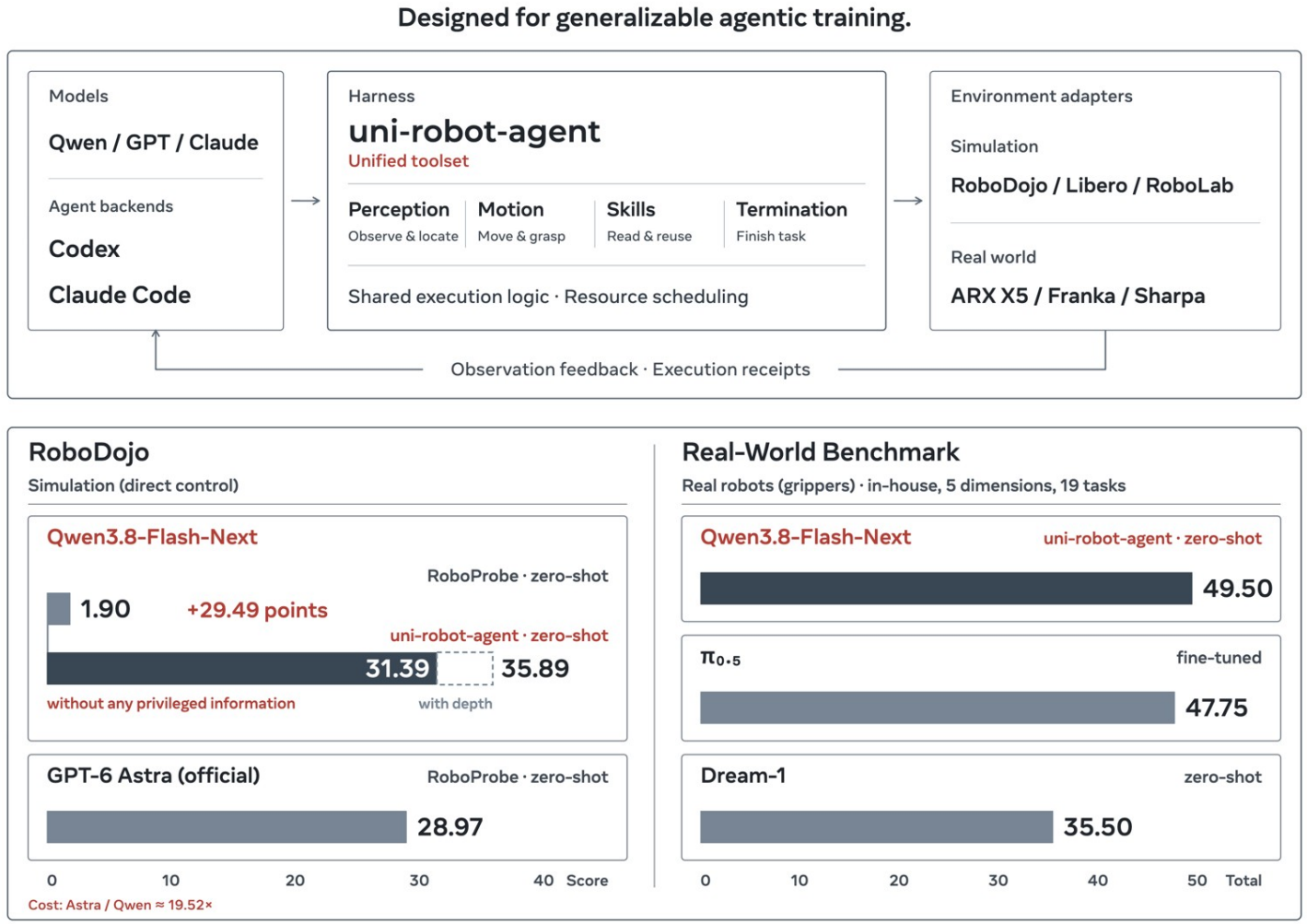
Prior work relies on simulator ground truth, task-success feedback, or task-specific skills

Authors' position

Build a general, effective robot-use harness before blaming the model

- Systematically understand and maximize open models' abilities
- Use only info available in real deployment
- Unify interaction format to accumulate training data
- Lay the groundwork for robot-oriented training

One unified toolset connects any model to any robot environment



Unified

Shared tool semantics, units and feedback; logs can be merged for training

Efficient

Inference overlaps execution; same-resource actions stay ordered

Realistic

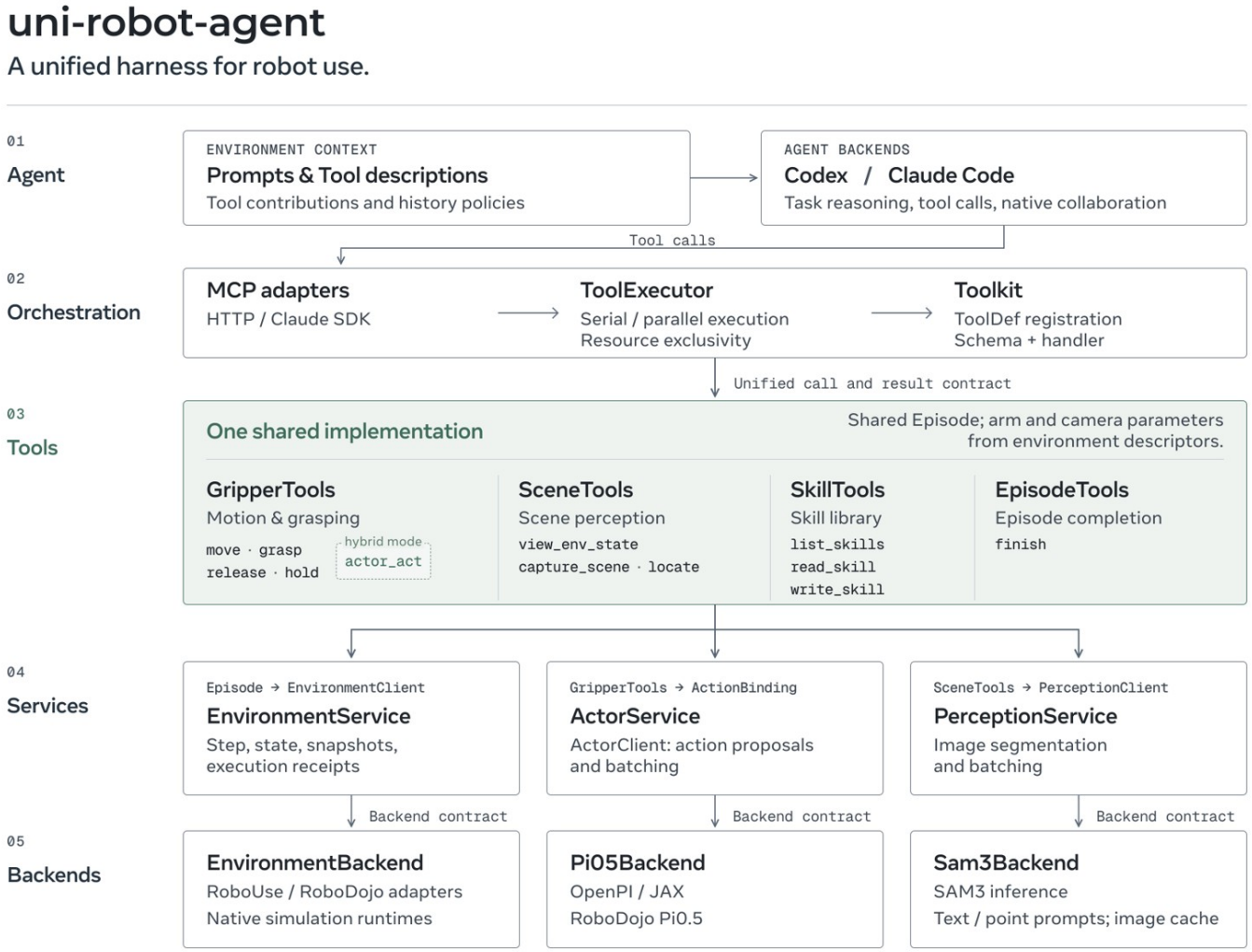
Only RGB, calibration, proprioception, robot description, instruction

Modular

Runtime, tools, perception and backends are swappable

Source: Fig. 1

Five-layer modular architecture: clear interfaces, swappable parts



01 Agent

Prompts, tool descriptions, context policy; Codex / Claude Code

02 Orchestration

MCP adapter + ToolExecutor: serial/parallel, exclusive resources

03 Tools

move / grasp / locate / skills / finish: one shared implementation

04 Services

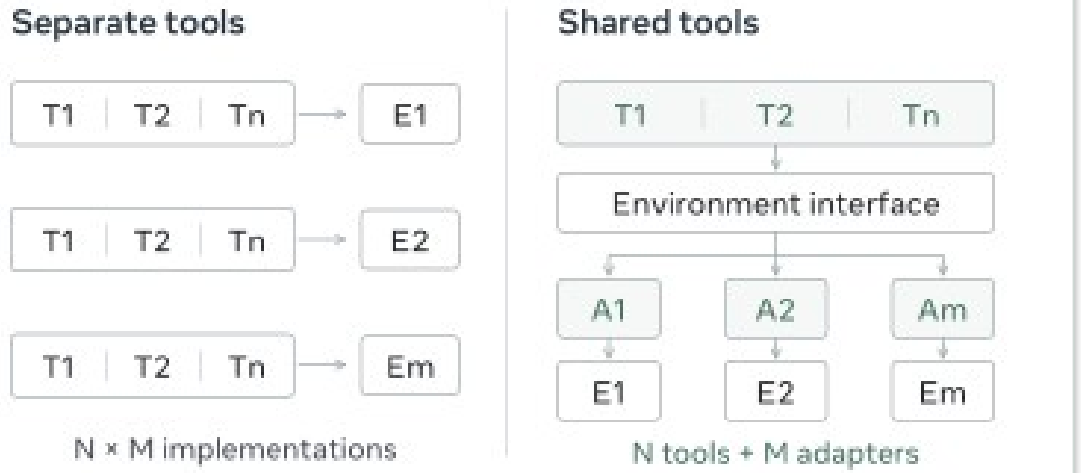
Environment, action model (π0.5), perception (segmentation) services

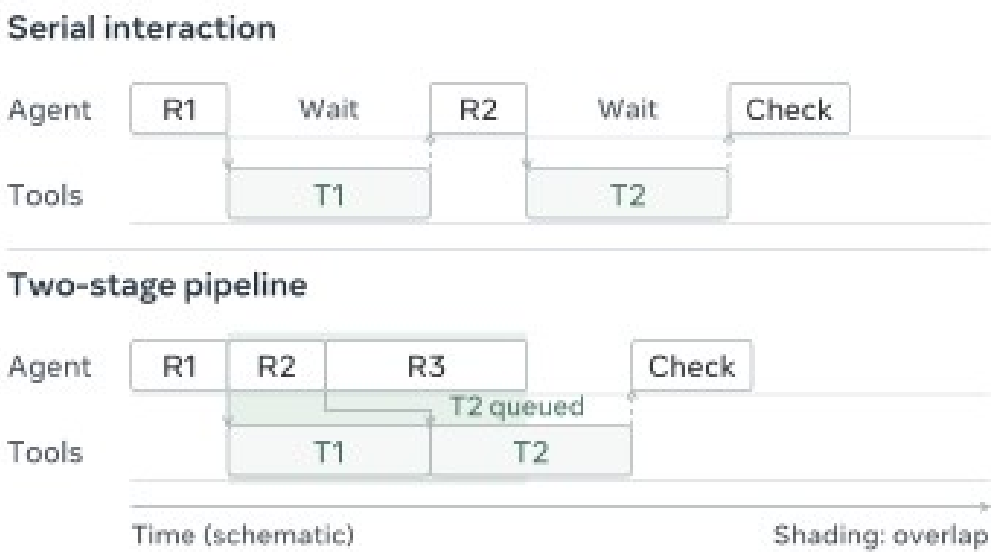
05 Backends

RoboDojo adapter, OpenPI/JAX, SAM3

Source: Fig. 2

Shared tools end N×M work; 2-stage pipeline overlaps inference/action



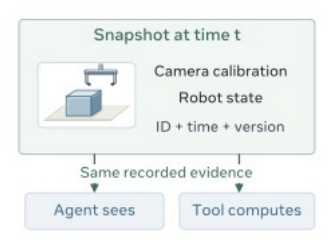


Unified interface The move tool generates subgoals and monitors progress; adapters only translate to native commands. New environment = one adapter + config

Two-level concurrency Agent–tool: background calls return a job handle; tool–tool: exclusive resources run in submission order, cancelable via flush_queue

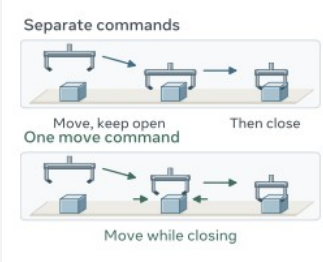
Source: Fig. 3, Fig. 4

Six engineering details that make interaction reliable



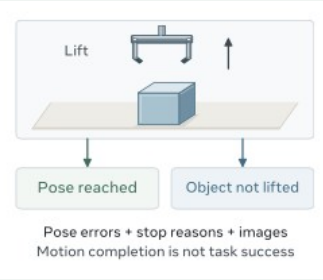
(a) Single snapshot

Images, calibration and robot state bound in an immutable snapshot



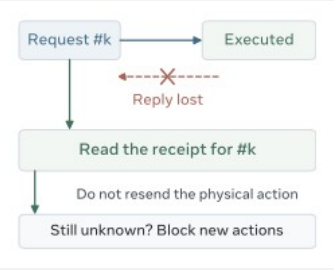
(c) Synced arm & gripper

One move sets target pose and gripper opening



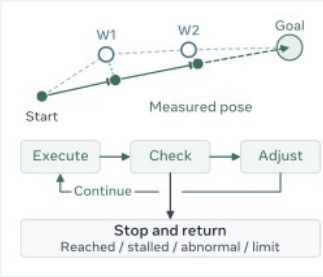
(e) Diagnostic feedback

Pose error + stop reason + image; reached ≠ success



(b) Check receipt before retry

Lost replies never resend motions; if unsure, block new ones



(d) Progress-based stop

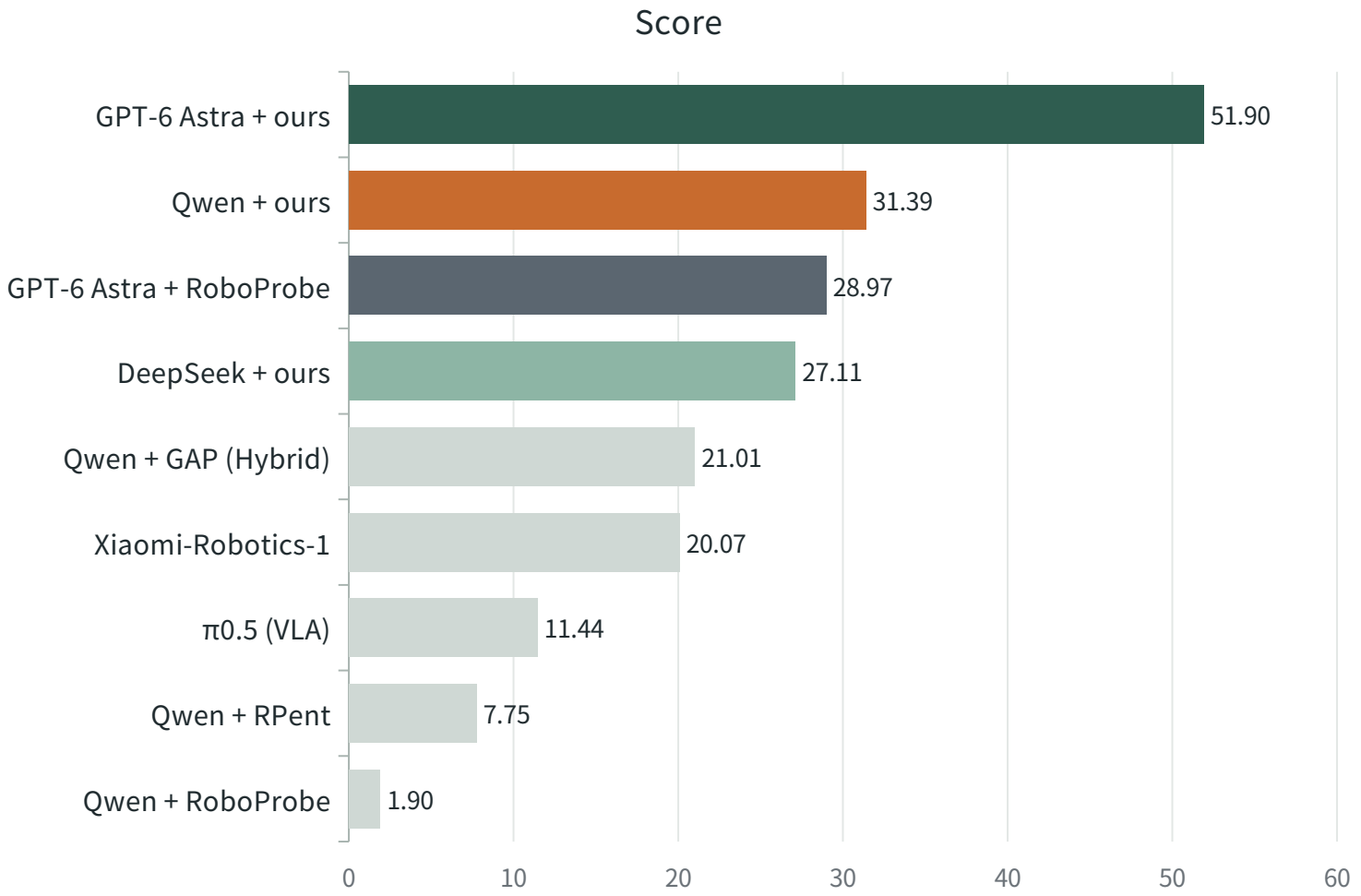
Waypoints from measured pose; returns on arrival, stall, fault or limit

(f) Keep useful context

Old images batch-replaced by placeholders for cache reuse

Source: Fig. 5 (a–f)

RoboDojo: open Qwen scores 31.39, beating official GPT-6 Astra (28.97)



+29.49

Same Qwen model: RoboProbe 1.90 → ours 31.39

+10.38

vs. strongest open harness GAP (Hybrid, with π0.5)

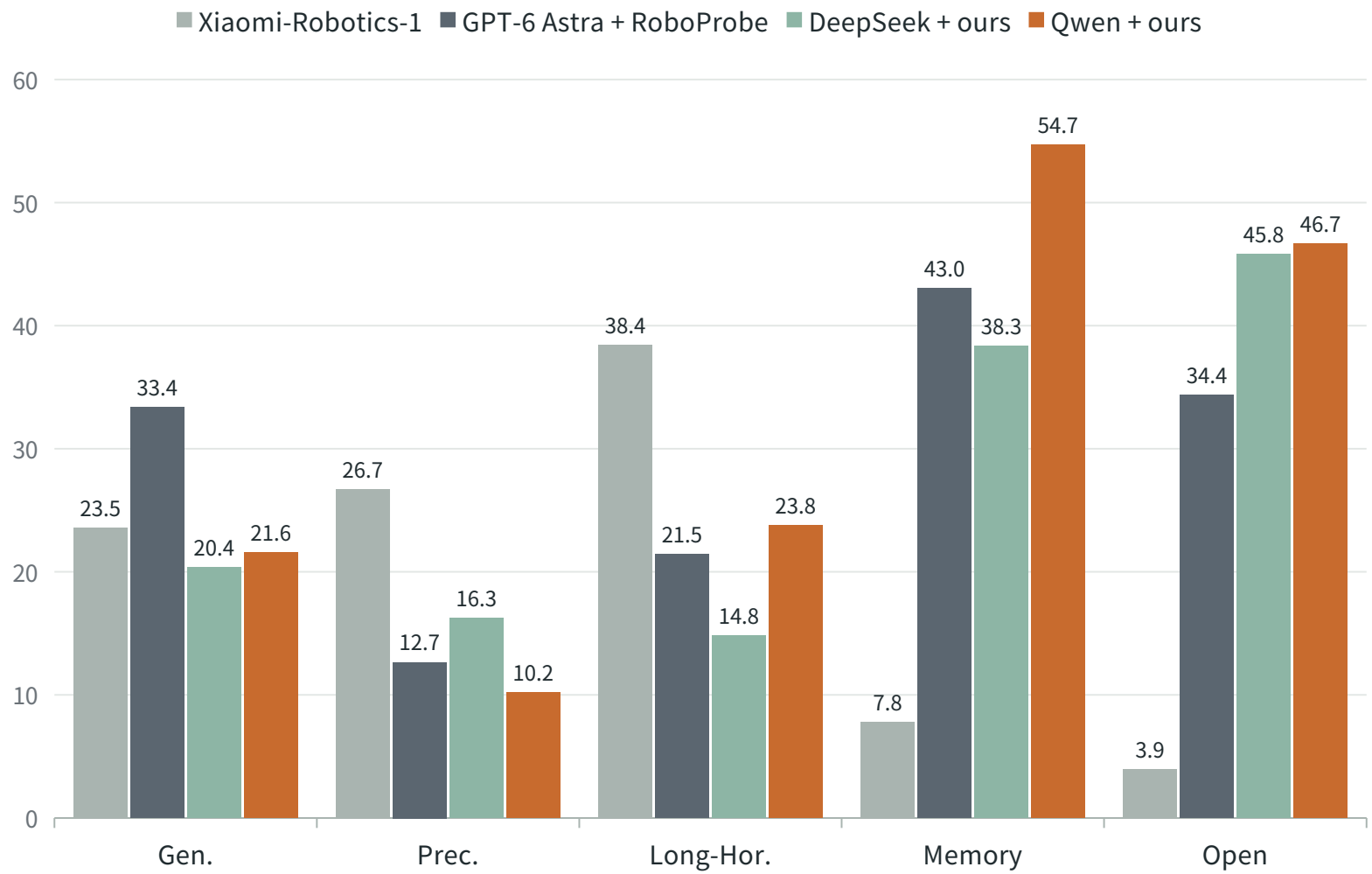
28.97 → 51.90

Closed GPT-6 Astra also gains greatly with ours

Setup: 42 tasks × 3 seeds × 2 layouts = 252 episodes; no depth, no task-checker feedback

Source: Table 1 (Score, 0–100)

Agents: strong memory/open vocab, weak precision/long horizon



Strong: memory / open vocab

Qwen 54.72 / 46.67, far above VLA/WAM (best under 14): general models' semantics and memory transfer directly

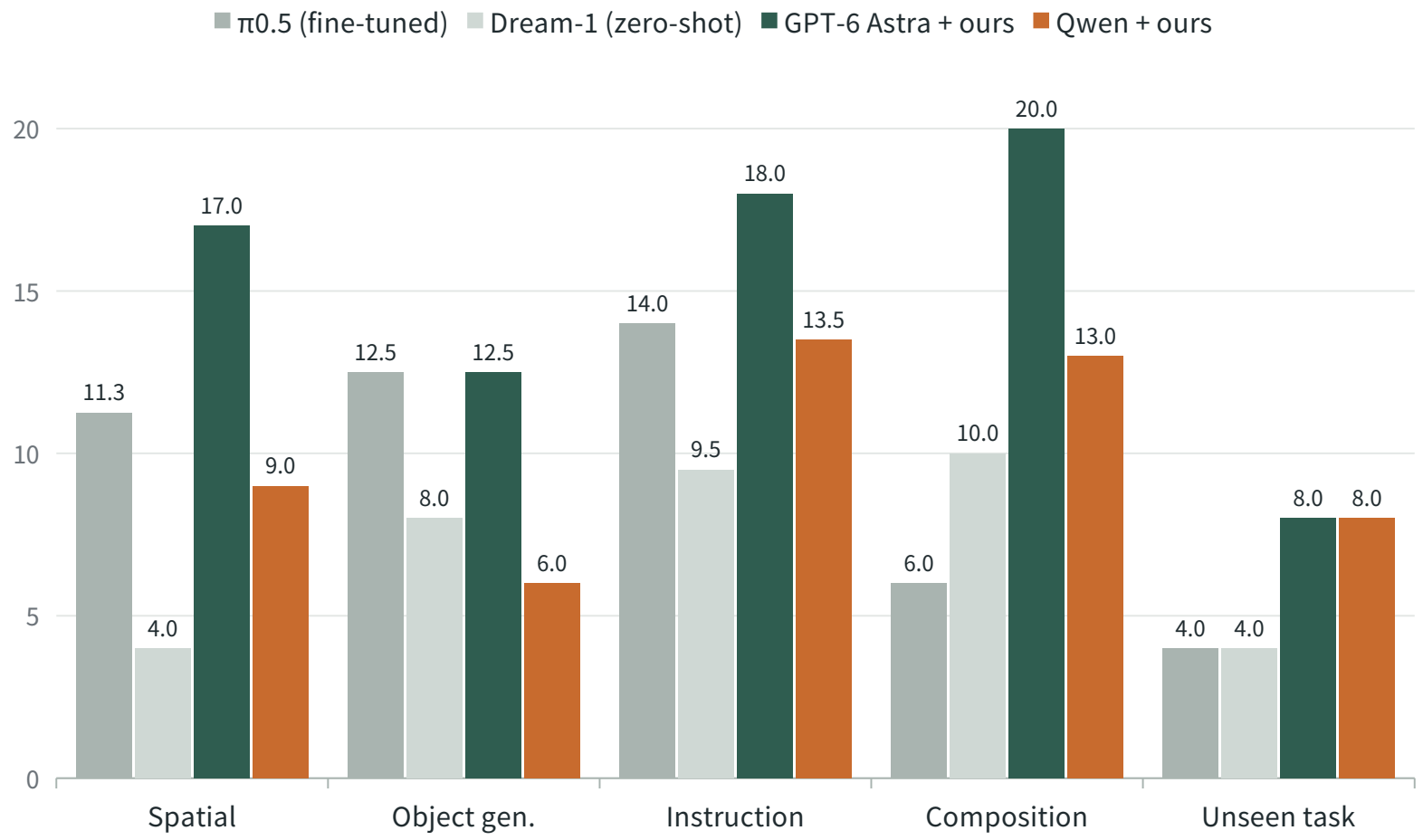
Weak: precision / long horizon

Qwen's precision success rate is only 2.08%; long-horizon 23.75 trails Xiaomi-Robotics-1's 38.39

Source: Table 1 (Score)

Experiments • Real Robot

19 real tasks: zero-shot Qwen matches fine-tuned π0.5; Astra hits 75.5



Agents: compositional/unseen; VLAs: object gen. (π0.5 new bowls 10/10, Qwen 2-arm handover 0/10)

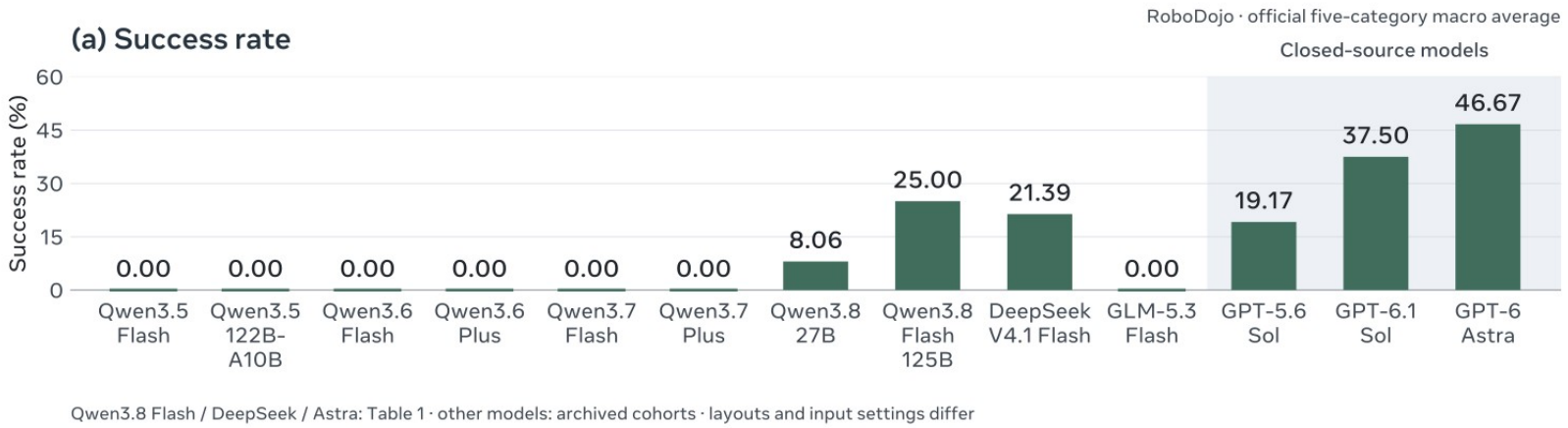
Total score (out of 100)

GPT-6 Astra	75.50
Qwen3.8-Flash-Next	49.50
π0.5 fine-tuned	47.75
Dream-1	35.50

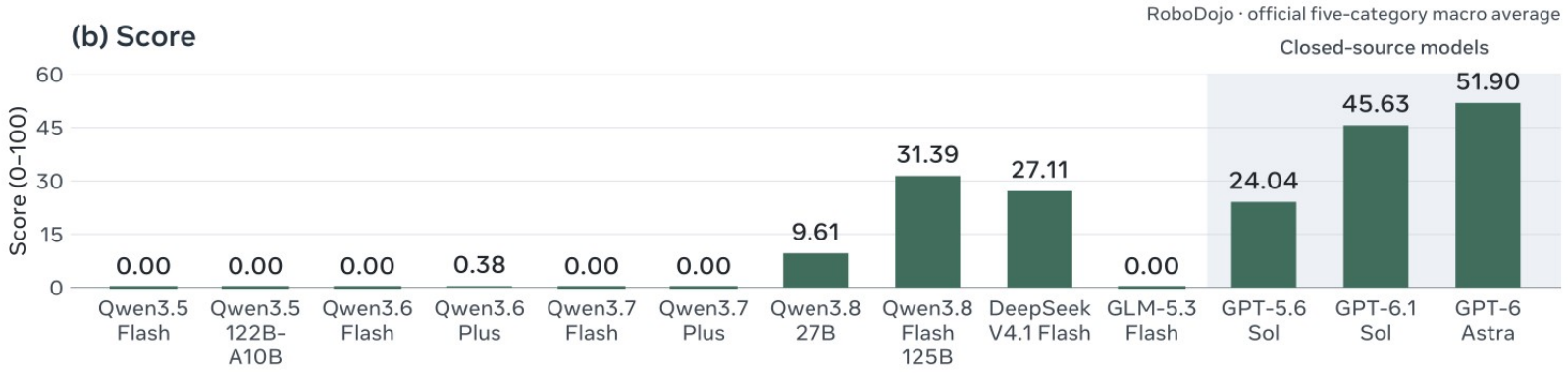
Note: on real robots our harness also uses D405 depth; π0.5 / Dream-1 use RGB only

Source: Table 2, Table 4, Appendix B

Scale isn't key: Qwen3.8-27B works; bigger old Qwens all get 0%



Qwen3.8 Flash / DeepSeek / Astra: Table 1 · other models: archived cohorts · layouts and input settings differ



Qwen3.8 Flash / DeepSeek / Astra: Table 1 · other models: archived cohorts · native partial credit retained

Older Qwen gens: all 0

Qwen3.5–3.7 Flash/Plus and Qwen3.5-122B: 0% success; GLM-5.3 Flash also 0

Smaller, newer models win

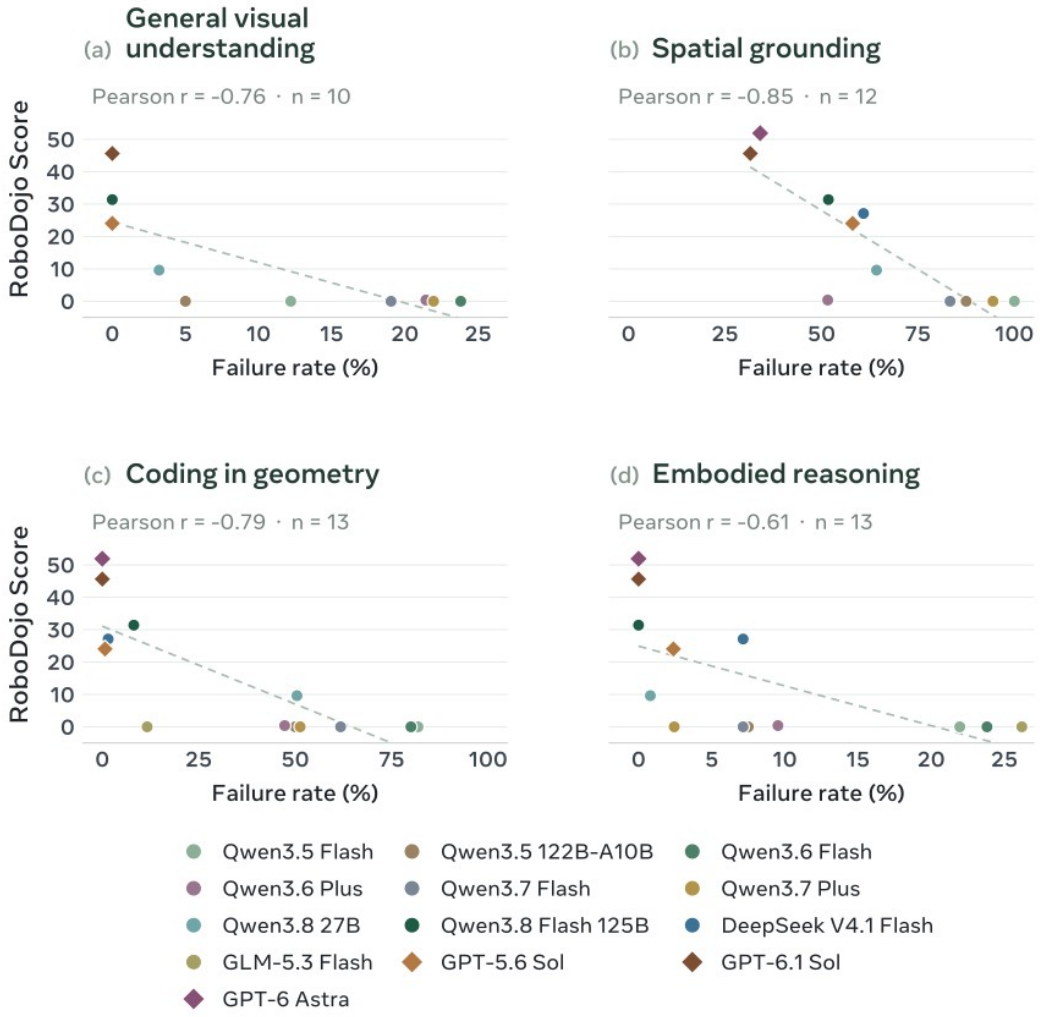
Qwen3.8-27B： 8.06%； Flash-Next（125B）： 25.00%

GPT series got there earlier

GPT-5.6 Sol 19.17% / 24.04； GPT-6.1 Sol 37.50% / 45.63

Source: Fig. 6

Four failure modes track score: visual, spatial, geometric, embodied



r = −0.76

Visual understanding

Reads 0 as 9, inverted cup as upright; early Qwen 12.2–23.8%

r = −0.85

Spatial grounding

Inconsistent multi-view points rejected; best GPT-6.1 Sol still 31.5%

r = −0.79

Geometric coding

Inverted transforms, mixed quaternion conventions; early Qwen 47.3–81.8%

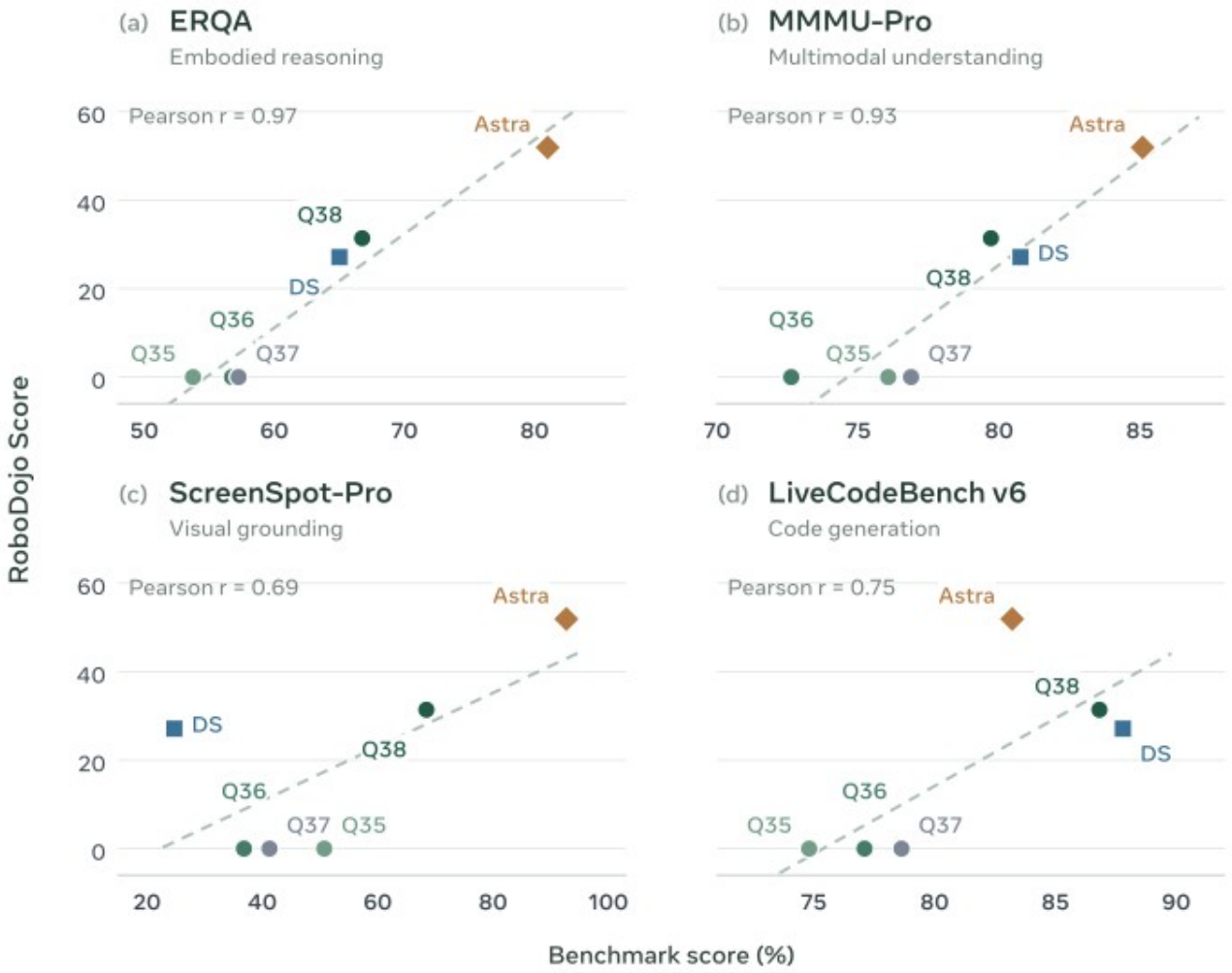
r = −0.61

Embodied reasoning

Implausible grasp poses; GLM 26.2% (judged by GPT-6 Astra)

Source: Fig. 7, § 4.3

Stronger general ability, better robot use: ERQA r=0.97



Pearson r (6 models)

ERQA
Embodied reasoning
0.97

MMMU-Pro
Visual understanding
0.93

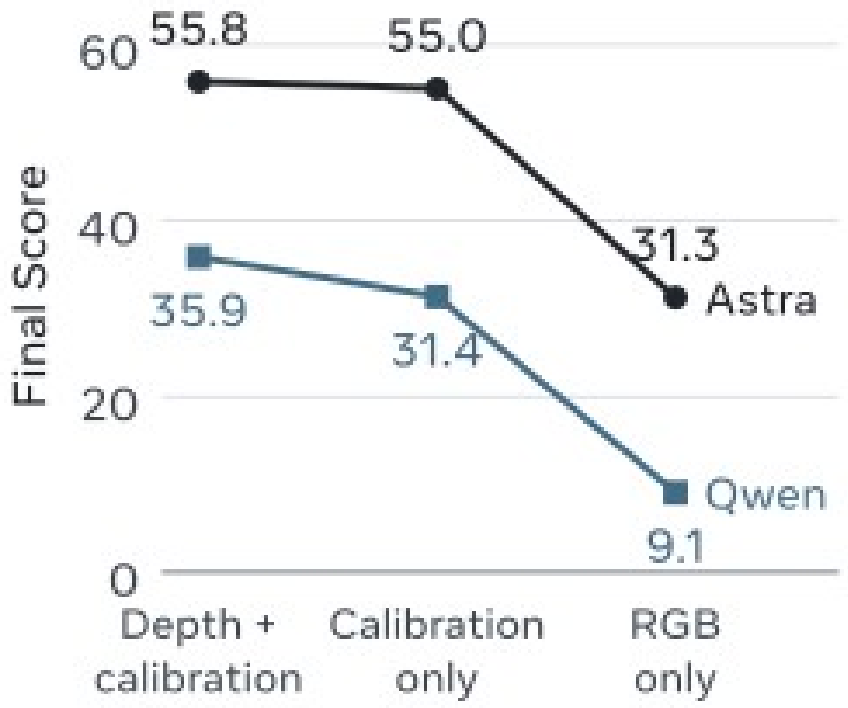
LiveCodeBench v6
Code generation
0.75

ScreenSpot-Pro
2D visual grounding
0.69

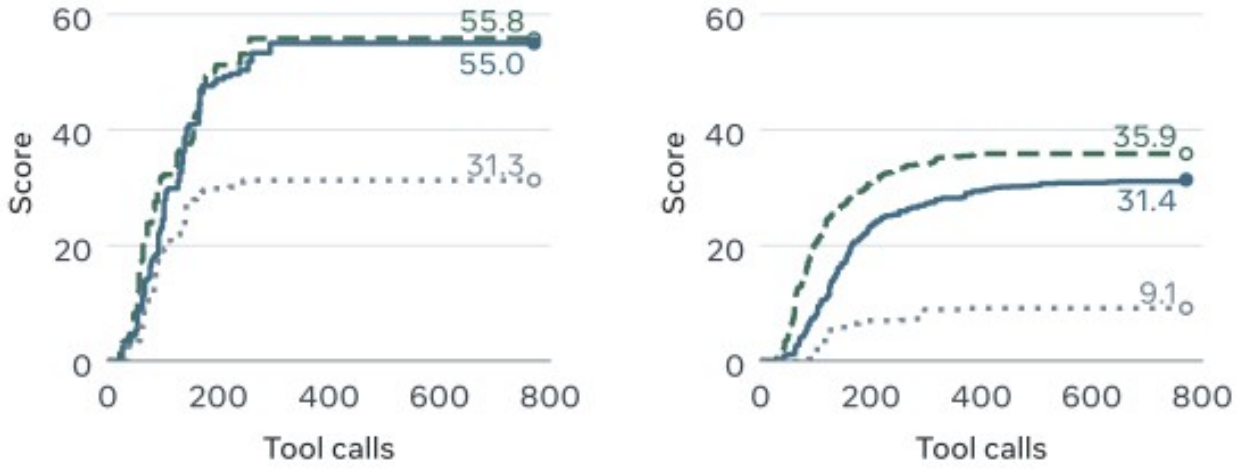
Takeaway: RUA ability composes general abilities; training should target each ability, not just fit action outputs

Source: Fig. 8

Calibration replaces depth; without it, Qwen falls to 9.1



Depth+calib / calib only / RGB only
Astra 55.8 / 55.0 / 31.3; Qwen 35.9 / 31.4 / 9.1



Score vs. tool calls: left GPT-6 Astra, right Qwen3.8-Flash-Next (dashed = depth, solid = calibration, dotted = RGB only)

98.6%
Astra retained w/o depth

87.5%
Qwen retained w/o depth

9.1
Qwen RGB-only score

Source: Fig. 9

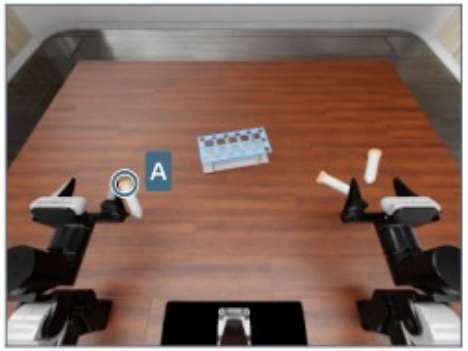
No depth? Qwen writes its own triangulation code for 3D positions

Recovering 3D positions from matched pixels

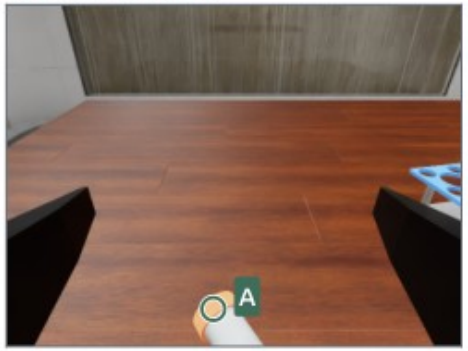
insert_tubes · Qwen3.8-Flash-Next · Camera intrinsics and extrinsics

01 Observe and reconstruct

Head camera



Left wrist camera



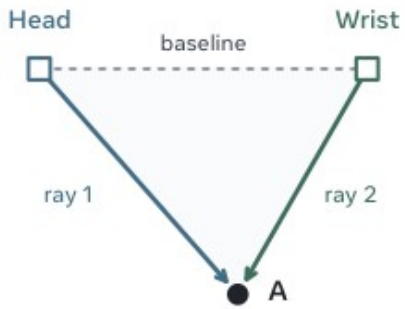
A marks the same tube cap in both views.

02 Execute generated code

```
def intersect_rays(pairs):
    A=np.zeros((3,3)); b=np.zeros(3)
    for cam,uv in pairs:
        o=center(cam); d=ray(cam,uv)
        M=np.eye(3)-np.outer(d,d)
        A+=M; b+=M@o
    return np.linalg.solve(A,b)
```

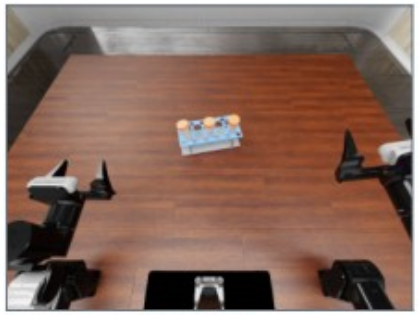
3D position (m)	(−0.3151, −0.1347, 0.7824)
Reprojection error	0.46 px / 1.94 px

Triangulation · schematic



Two views → one 3D estimate

03 Manipulate



Three tubes inserted
Final observation

insert_tubes: multi-view triangulation

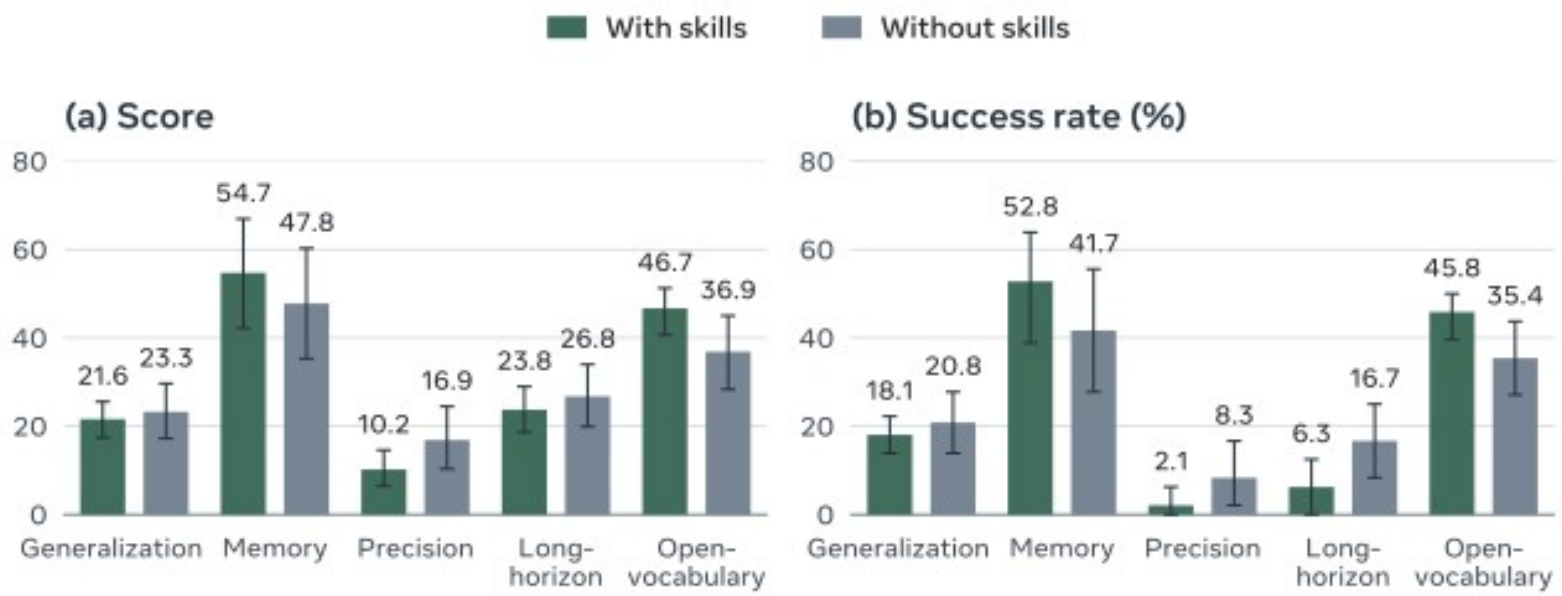
Self-written intersect_rays least-squares solver; cap 3D reprojection error 0.46 / 1.94 px; all three tubes inserted

cover_blocks: geometry hypotheses

Scans table heights 0.74–0.77 m; ray–plane intersection gives cup width 0.075–0.071 m, then covers and uncovers

Source: Fig. 15, Fig. 16 (Appendix E)

No skill library: 96.6% of score kept, no task-specific know-how



96.6%

No skills 30.31 pts / 24.58% SR
Skills 31.39 pts / 25.00% SR

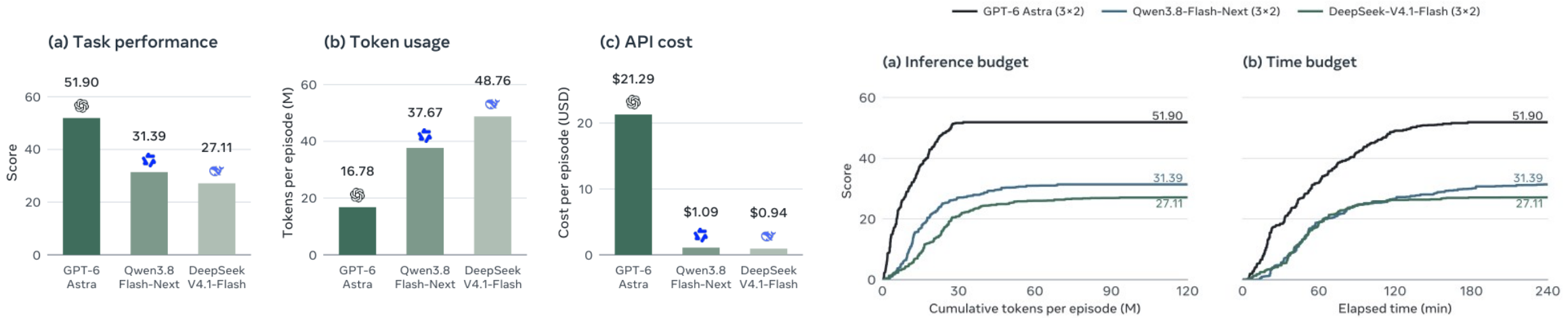
Uneven by dimension

Without skills: precision 10.2→16.9, long-horizon 23.8→26.8 rise; memory 54.7→47.8, open vocab 46.7→36.9 fall; wide error bars

Skill library and skill writing disabled; task instruction, environment description and standard robot tools kept; error bars ≈ 95% bootstrap CIs

Source: Fig. 10

Qwen reaches 60% of Astra's score at ~1/20 the cost



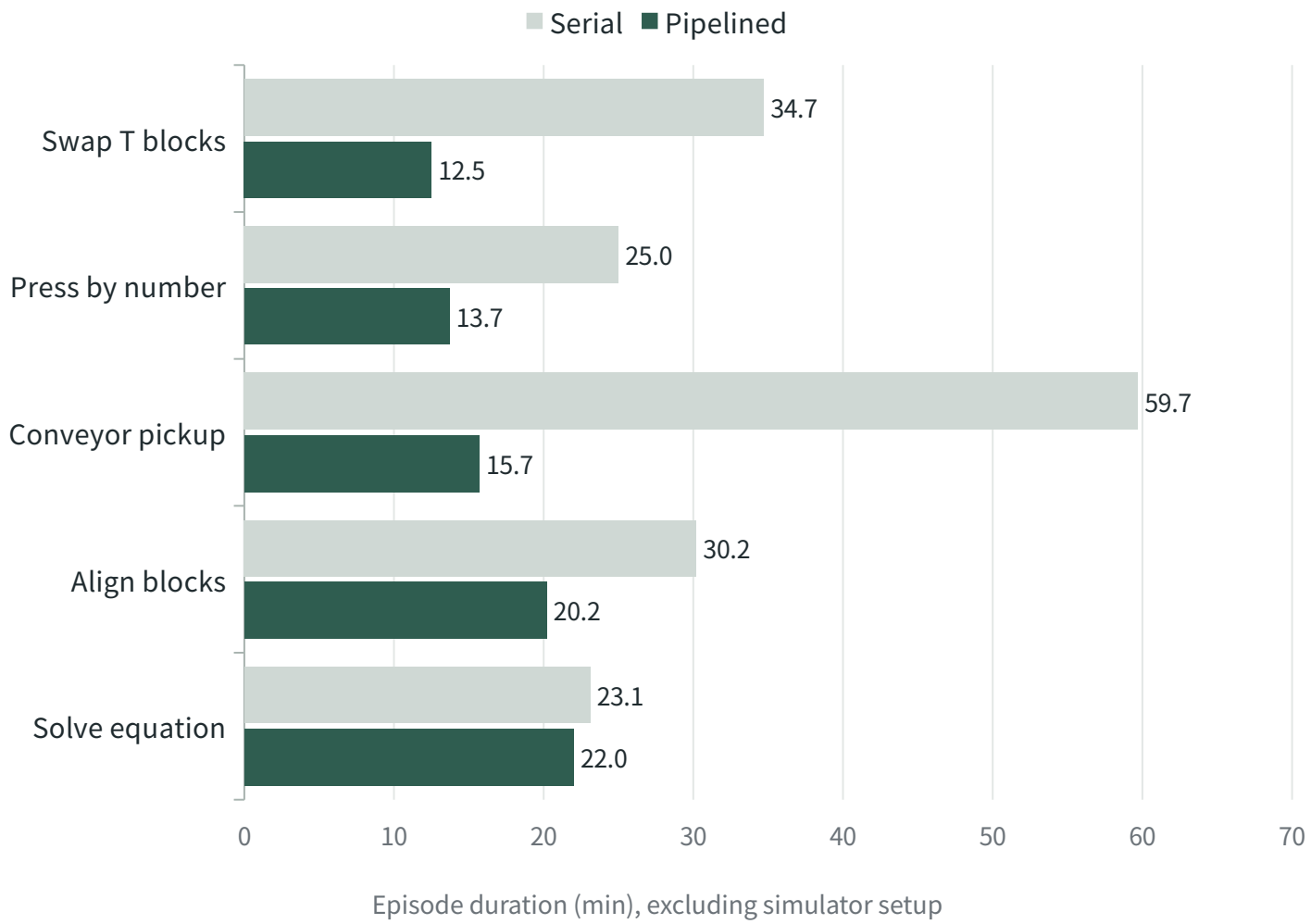
To reach 25 pts	GPT-6 Astra	Qwen3.8	DeepSeek
Token / episode	7.4M	23.5M	49.9M
Wall-clock	42.6 min	95.5 min	99.2 min
Cost/episode	$21.29	$1.09	$0.94

Reading: Qwen scores ~60% of Astra (31.39 / 51.90) at ~1/19.5 the per-episode cost, but Astra is far more token- and time-efficient

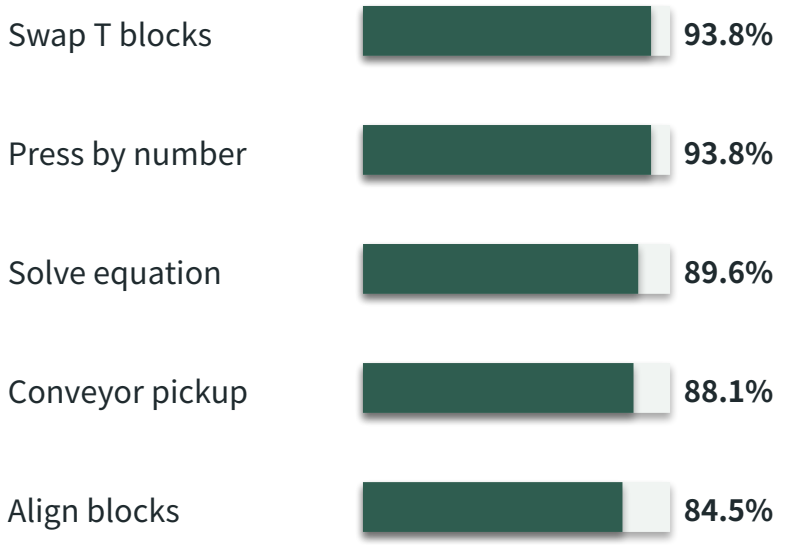
Source: Fig. 11, Fig. 12, Appendix C

Analysis • Execution Efficiency

Pipelined execution: episodes up to 73.7% shorter



Action time overlapped with inference



What happens during overlap?

E.g., computing clearance for a lower grasp while releasing; preparing 90°/180° rotation candidates in transit (Appendix F)

Source: Fig. 13, Fig. 14 (GPT-6 Astra, 1 run per config)

Critique

Verdict: solid harness; fairness and statistical rigor need work

Strengths

- First system to beat official closed-model RoboDojo results with open models, without privileged info or closed-model distillation
- Same model, new harness: 1.90 → 31.39; many "open models fail" claims may be harness issues
- Consistent snapshots, idempotent requests, progress-based stops, cache-friendly context: directly reusable on real robots
- Four abilities get measurable metrics, cross-checked with general benchmarks, pointing to training directions

Limitations / caveats

- Real-robot comparison not fully fair: ours uses D405 depth, baselines RGB only; 49.50 vs 47.75 is a small gap
- Inconsistent setups: generation comparison and Astra depth ablation use 1 seed × 1 layout; correlations use only 6–13 models; embodied reasoning judged by Astra
- Precision (Qwen SR 2.08%) and long horizon remain weak; ~38M tokens per episode
- "Groundwork for training" not yet delivered: no training experiments

Source: whole paper

Three Takeaways

01

Harness before model

A good harness (unified tools, diagnostic feedback, calibration) lifts open models from "near 0" to beating official closed-model results

02

Composite abilities

Visual understanding, spatial grounding, geometric coding and embodied reasoning jointly determine robot use; train them jointly, not just fit actions

03

Open, cheap, trainable

~1/20 of Astra's cost and locally deployable; unified interaction logs pave the way for training on robot experience

Open issues: fine manipulation, reliable long-horizon execution, inference speed and token cost, and real training experiments